EVERY CONNECTED SEMISIMPLE LIE GROUP WITH FINITE CENTRE HAS A LATTICE WHOSE FINITE-DIMENSIONAL REPRESENTATIONS ARE INTEGRAL

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Allowed standard prerequisites (editorial).

- The arithmetic lattice construction: for an adjoint Q-group the set of g with Ad g integral is a lattice. Used inside the counted proof at source line 365 and attributed by the author to the cited book at Theorem 4.14, page 220. Provides the subgroup whose two properties the statement asserts.
- Existence of invariant Z-lattices for integral representations. Used inside the counted proof at source line 372 and attributed by the author to a remark on page 173 of the same cited book. Turns the Q-defined representation into a rho(Gamma)-invariant Z-lattice.
- The existence of an R-universal Q-form for a real semisimple Lie algebra. The result the paper proves; used at the start of the counted proof at source line 362. Supplies the Q-form in terms of which the lattice is defined and the rationality of the differential of rho.

Corollary 0.1. *Let G be a connected, semisimple Lie group with finite center. Then there is a discrete subgroup Γ of G, such that*

(1) *G/Γ has finite volume (so Γ is a "lattice" in G), and*

Date: doi:10.3842/SIGMA.2015.034; CC BY 4.0.

(2) *if* $\rho\colon G \to \mathrm{GL}(n,\mathbb{R})$ *is any finite-dimensional representation of* G*, then* $\rho(\Gamma)$ *is conjugate to a subgroup of* $\mathrm{GL}(n,\mathbb{Z})$.

Proof of Corollary 0.1. Choose an $\mathbb{R}$-universal $\mathbb{Q}$-form $\mathfrak{g}_{\mathbb{Q}}$ of the Lie algebra $\mathfrak{g}$ of G, let $\overline{\mathbf{G}}$ be the corresponding adjoint $\mathbb{Q}$-group, and let $\Gamma = \{g \in G \mid \operatorname{Ad} g \in \overline{\mathbf{G}}(\mathbb{Z})\}$, so Γ is a lattice in G (cf. [7, Theorem 4.14, p. 220]).

Suppose $\rho\colon G \to \mathrm{GL}(n,\mathbb{R})$ is a finite-dimensional representation of G. By replacing ρ with a conjugate, we may assume $d\rho(\mathfrak{g}_{\mathbb{Q}}) \subseteq \mathfrak{gl}(n,\mathbb{Q})$ (because $\mathfrak{g}_{\mathbb{Q}}$ is $\mathbb{R}$-universal). Then the corresponding representation $\widetilde{\rho}$ of the universal cover $\mathbf{G}$ of $\overline{\mathbf{G}}$ is defined over $\mathbb{Q}$, so there is a $\widetilde{\rho}\big(\mathbf{G}(\mathbb{Z})\big)$-invariant $\mathbb{Z}$-lattice in $\mathbb{Q}^n$ [7, Remark on p. 173]. Then, since $\widetilde{\rho}\big(\mathbf{G}(\mathbb{Z})\big)$ contains a finite-index subgroup of $\rho(\Gamma)$, there is also a $\rho(\Gamma)$-invariant $\mathbb{Z}$-lattice in $\mathbb{Q}^n$, so $\rho(\Gamma)$ is conjugate to a subgroup of $\mathrm{GL}(n,\mathbb{Z})$. □

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